KIEU STUDIO · GITHUB RADAR · EPISODE 6 · CHEAT SHEET

text-to-cad

Design physical products with AI.

github.com/earthtojake/text-to-cad

+7,976	24,386	MIT
stars in 30 days · ≈ 266 stars/day	total stars · GitHub API	license · created Apr 22, 2026

A plugin/skills set for coding agents: the agent writes each 3D model as a Python script (build123d on Open CASCADE), builds STEP, GLB, STL or 3MF files, checks manufacturability, makes engineering drawings and connects to 3D printing, sheet metal and CNC services.

How it works in six steps

1. **Describe it** — in words or an image
2. **Python script** — build123d on Open CASCADE
3. **cadgen engine** — runs python model.py, builds an .xbf document in memory
4. **Export files** — STEP, GLB, STL, 3MF
5. **Check it** — DfM, DfAM, engineering drawings
6. **CAD viewer** — refreshes after every rebuild

Quick start

```
# macOS / Linux
curl -fsSL https://texttocad.dev/install.sh | sh

# Windows (PowerShell)
irm https://www.texttocad.dev/install.ps1 | iex

# Diagnose problems
cadgen doctor

# Use it
Ask your agent for a part or an assembly: it writes the
script, runs it and shows the model

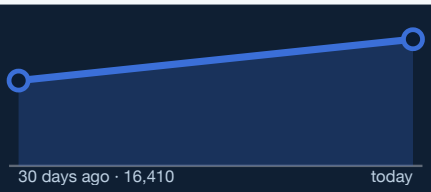
# Needs: Installer sets up uv if missing; works with Claude Code, Codex,
Cursor, Gemini, Grok
```

Example prompt: (illustration) Design a 3D-printable phone stand with walls at least two millimeters thick.

Key commands / features

- **`CAD`** — create and edit models from words or images
- **`step.parts`** — find off-the-shelf screws, bearings, motors, connectors (STEP)
- **`Engineering Drawing`** — dimensioned PDF drawings: views, hole callouts
- **`DXF`** — 2D profiles, gaskets, cut layouts
- **`DfAM Check / DFM`** — printability, sheet metal, CNC, injection molding

Growth



Two real points: the start of the window and today, axis from zero. The gain is as reported by GitHub Trending; we did not rebuild the star history.

Why this repo exists

- AI is used to words and code, but a real object needs an accurate 3D model, manufacturability checks and the right file formats.
- The model is code, so it can be edited, compared and versioned.

What to use it for

- Custom 3D-printed parts (stands, cases, mounts)
- Laser-cut or CNC parts from DXF
- Check manufacturability before sending to a shop
- Robot descriptions for simulation

Think twice before you use it

- The installer runs a script from the web and configures every agent app on your machine: read it first.
- By default it sends usage stats and crash reports with a random ID, never files, paths or prompts (README); the opt-out is in the privacy policy.
- Always check the model and tolerances before manufacturing or ordering parts.

Sources: the repo's README, https://github.com/earthtojake/text-to-cad, https://www.texttocad.dev. Stars: GitHub API; gains: GitHub Trending (as reported by Trending), snapshot Oct 31, 2026; stars measure attention, not quality: re-check before relying on them. Summarized in Kieu Studio's own words; commands copied from the README; no third-party logos or screenshots.
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